Web Applications Development

Mid Term Examination Solutions (E1PY204B)

Section A: Web Communication & Structures

1. List the steps involved in web communication when a webpage loads.

The process involves several key steps:

1. **DNS Lookup:** The browser converts the URL into an IP address.
2. **TCP Handshake:** A connection is established between the client and the server.
3. **HTTP Request:** The browser sends a request for the webpage resources.
4. **Server Processing:** The server handles the request and sends back an HTTP response.
5. **Browser Rendering:** The browser parses HTML, CSS, and JS to display the page.

3. Explain fieldset and legend usage.

The `<fieldset>` tag is used to group related elements within a web form, creating a visual box around them. The `<legend>` tag provides a caption or title for the fieldset, improving accessibility and form organization.

Section B: CSS & Styling Logic

5. Apply CSS cascade priority rules to determine the final style.

The final style of an element is determined by the following hierarchy (lowest to highest priority):

- **External CSS:** Linked via `<link>` tags.
- **Internal CSS:** Defined within `<style>` tags in the head.
- **Inline CSS:** Defined directly within the element's `style` attribute.
- **Specificity:** Calculated based on selectors (ID > Class > Element).
- **!important:** Overrides all other declarations.

Section C: JavaScript Logic

6. Outline the scope and hoisting in JavaScript.

- **Scope:** Determines the accessibility of variables (Global, Function, or Block scope using `let/const`).
- **Hoisting:** JavaScript's behavior of moving declarations to the top of the current scope. Function declarations are fully hoisted, while `var` is hoisted with a value of `undefined`. `let` and `const` are hoisted but remain in a "Temporal Dead Zone."

7. JavaScript Palindrome Check Program.

```
function checkPalindrome(str) {
    const cleanStr = str.toLowerCase().replace(/[^a-z0-9]/g, '');
    const reversedStr = cleanStr.split('').reverse().join('');

    if (cleanStr === reversedStr) {
        console.log("The string is a palindrome.");
    } else {
        console.log("The string is not a palindrome.");
    }
}

checkPalindrome("Level"); // Output: The string is a palindrome.
```